

Assumption University
Vincent Mary School of Science and Technology
CSX3007 Quiz1 2/2023 Total 64 marks **KEY**

Instructions: Read the questions carefully and answer each question in the spaces provided on the examination paper (*please do not use pencils for your answer*). Students are not allowed to use an electronic calculator. This is a closed-book examination.

1. 20 marks) **Number System**

- 1.1) (1 mark) What is the largest unsigned number that can be stored in a 32-bit register?
11111111 11111111 11111111 11111111 OR 0xFFFFFFFF
- 1.2) (1 mark) Show the 8-bit sign-magnitude representation of -9. **10001001**
- 1.3) (2 marks) Find the decimal equivalent of the following eight-bit 2's complement binary numbers:
- 1.3.1. (1 mark) 11111011_2 **-5**
- 1.3.2. (1 mark) 11110110_2 **-10**
- 1.4) (2 marks) Convert the following unsigned binary numbers to decimal:
- 1.4.1. (1 mark) 01000111_2 **71**
- 1.4.2. (1 mark) 10001000_2 **136**
- 1.5) (1 mark) Convert the following unsigned binary number to hexadecimal:
 110101011010110_2 **6AD6H**
- 1.6) (1 mark) Convert the following hexadecimal number to decimal: AE_{16} **174**
- 1.7) (1 mark) Repeat **question 1.6**, but convert it to an *unsigned binary* number. **10101110**
- 1.8) (1 mark) Convert the following hexadecimal number into *two's complement binary* number: $8EC_{16}$ **0111000100111110**
- 1.9) (3 marks) Answer the following:
- 1.9.1. (0.5 marks) Convert 2857_{10} to BCD (Binary Coded Decimal).
0010100001010111
- 1.9.2. (0.5 marks) Convert 110010011100111001_{BCD} to decimal. **32739**
- 1.9.3. (2 marks) Show the following decimal addition in BCD: $98 + 75 = 173$

$$\begin{array}{r} 1001\ 1000\ + \\ 0111\ 0101 \\ \hline 1\ 0000\ 1101\ + \\ 0000\ 0110 \\ \hline 1\ 0001\ 0011\ + \\ 0110\ 0000 \\ \hline 0001\ 0111\ 0011 \\ 1\quad\quad 7\quad\quad 3 \end{array}$$

- 1.10) (2 marks) Show the one-bit *shift right* and *rotate right* operations on the following 8-bit binary number: 10111010_2 **SR = 01011101** **RR = 01011101**
- 1.11) (2 marks) Check whether the following binary addition causes the arithmetic flags such as *carry*, *sign*, *parity*, and *overflow*: $10101011_2 + 10010100_2 = 00111111$, **CF = 1**, **OF = 1**, **SF = 0**
- 1.12) (3 marks) Convert -1.2010×2^{-16} into a 32-bit **IEEE** floating-point format.
1 (sign bit) 01101111 (8-bit biased exponent) 001100.....(23-bit mantissa)

2. (30 marks) **Combinational Logic**

2.1) (6 marks) Consider the following truth table (where A , B , and C are three input variables and Y is an output variable):

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

2.1.1. (2 marks) Write a Boolean equation in *sum-of-products* canonical form.

$Y = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC$

2.1.2. (4 marks) Minimize the equation from 2.1.1 using a *K-map* and sketch its combinational logic circuit. $\overline{A}\overline{B} + A = \overline{B} + A$

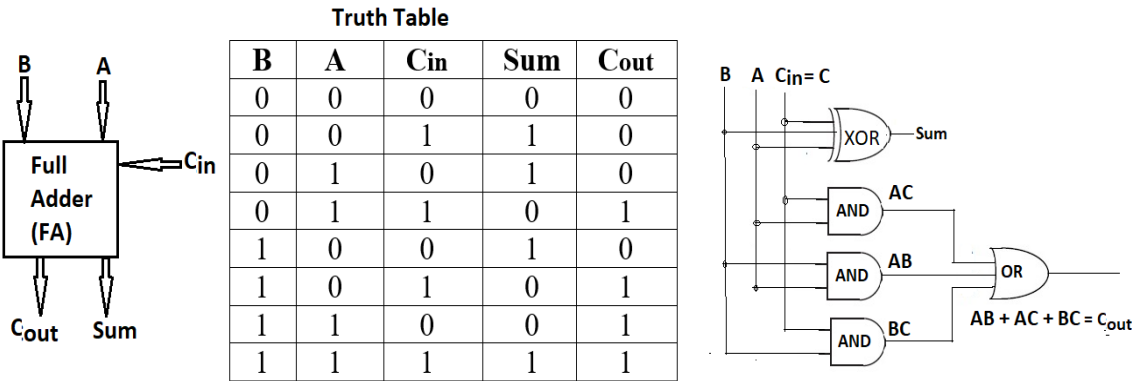
2.2) (4 marks) Simplify the following logic equations and *sketch the circuits*:

2.2.1. (2 marks) $Y = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + \overline{A}B = \overline{A}B + \overline{A}\overline{B} = A \text{ XOR } B$

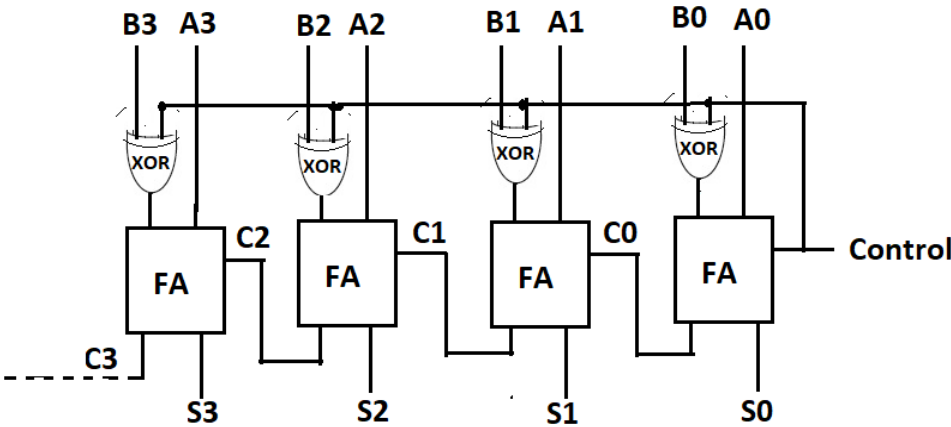
2.2.2. (2 marks) $Y = (A + \overline{B} + C) + (\overline{B}C) + ABC$

$\overline{A}B\overline{C} + B + \overline{C} + ABC = B + \overline{C}$

2.3) (5 marks) Design a full adder circuit using its truth table. Show how the full adders can perform binary addition and subtraction with the help of the circuit symbol of full adders (use the 4-bit binary values 1011_2 and 1001_2 to demonstrate a 4-bit addition and subtraction).



Full adders for addition and subtraction:



If **Control = 1**, the circuit act as a **subtractor**. If **Control = 0**, the circuit act as an **adder** (check this with the given 4-bit values).

- 2.4) (3 marks) Two 2-input logic gate functions are shown in **Figure 1** (a) and (b) (where A , B , and Y are inputs and output, respectively). Identify the logic gates.

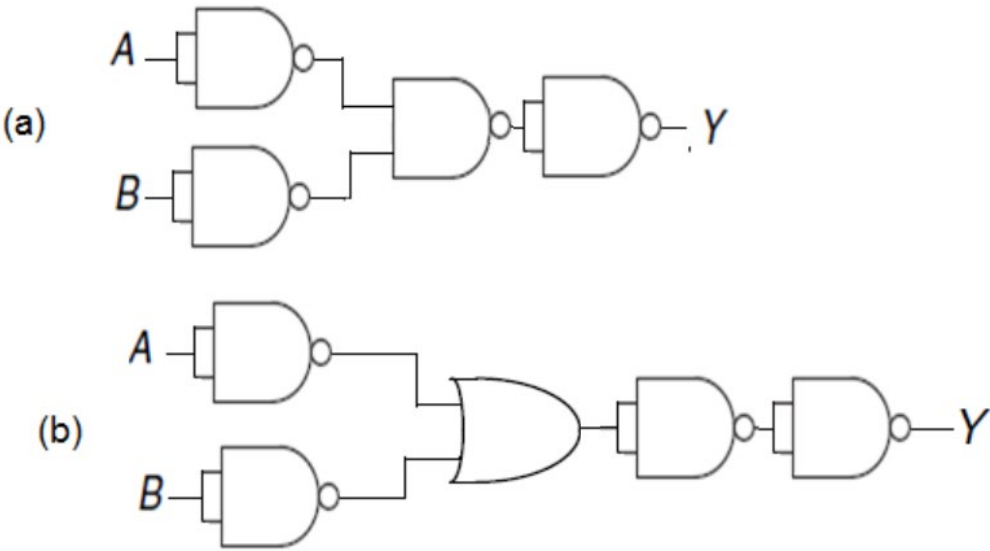


Figure 1.

Figure1(a) = $\bar{A} \bar{B} = \overline{A + B}$ (NOR), and Figure1(b) = $\bar{A} + \bar{B} = \overline{AB}$ (NAND)

- 2.5) (2 marks) Identify the circuit schematic shown in **Figure 2** by applying its input and output signal labels (where A , B , and C are inputs and X and Y are outputs).

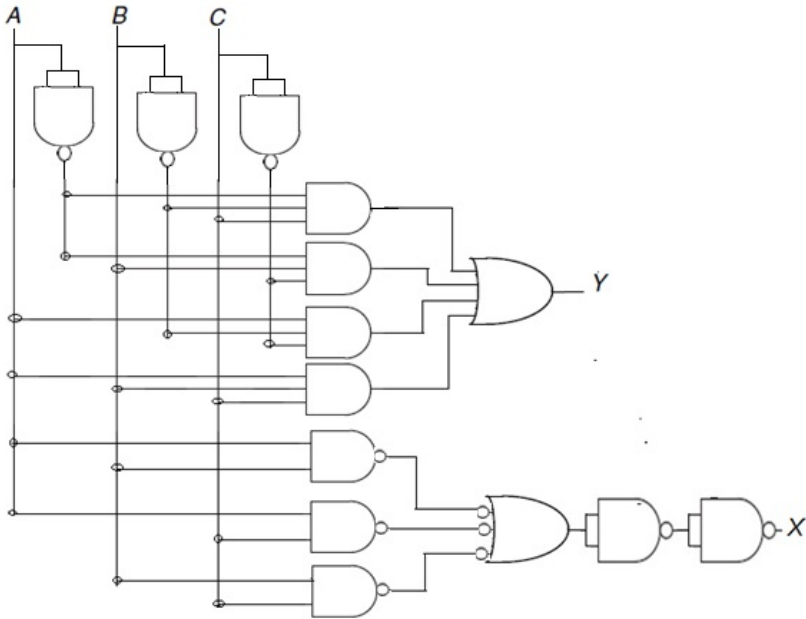


Figure 2.

This is a full adder circuit; A , B , and C are inputs, $Y = \text{sum}$, and $X = C_{out}$.

- 2.6) (4 marks) Create the circuit schematic of a **4-line Demultiplexer (DMUX)** using its truth table. Show how to convert the **DMUX** into a **4-line Decoder** and its circuit schematic.
Check my DUMX and Decoder Description
- 2.7) (4 marks) Show how a 4-line Decoder can select **four 16Kbyte (16K x 8)** memory chips. Each chip has a *chip selection* (CS) input; the chip is selected if CS is one means.
Each memory has 14 address bits (A0 to A13), and the decoder has two control inputs (A14 and A15) to select each memory chip (connected via their CS). The decoder and the memory chips pics are not shown here!
- 2.8) (3 marks) Assume that a computer system has a **4Gbyte** main memory (organization: **4G x 8**). Answer the following based on the main memory:
- 2.8.1. (1.5 marks) How many control input bits are in the address decoder of the memory? **32**
- 2.8.2. (1.5 marks) How many output bits are in the address decoder of the memory?
 2^{32} or 4G

3. (14 marks) Sequential Logic

- 3.1) (2 marks) Is it possible to design Static RAM (SRAM) with capacitors? Describe your answer. **This is not possible because SRAM is the fastest memory for cache memory applications. If it is made of capacitors, it has no use (its speed will be down due to its charge leakage and refreshing issues).**
- 3.2) (3 marks) Show the circuit schematics of an S-R latch using NOR gates. Explain how it can convert into a D-latch. **Check lecture slide for its figure and details**
- 3.3) (3 marks) Show a D Flip-flop's circuit schematic and how to create a 4-bit parallel-in-parallel-out register. **Check lecture slide for its figure and details**
- 3.4) (3 marks) Identify the name and function of the sequential logic circuit shown in **Figure 3** by correctly labeling its inputs and outputs. **JK latch, 1 = K, 2 = CLK, 3 = J, 4 = R, 5 = S, 6 = Q, 7 = $\bar{Q}$**

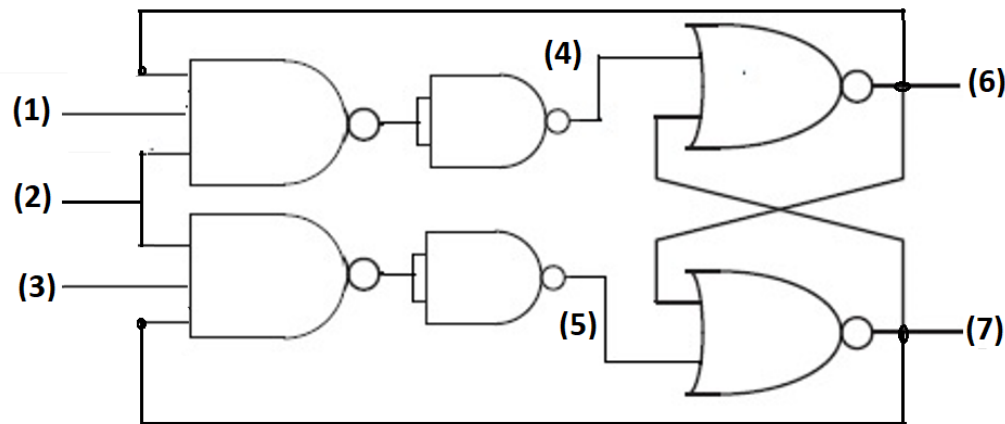


Figure 3.

- 3.5) (3 marks) Identify the sequential logic circuit shown in **Figure 4** and briefly describe its function by creating its truth table (where *Clk* is the clock input and Q_0 , Q_1 , Q_2 , and Q_3 are outputs). **Mod 12 asynchronous counter (it counts from 0000 (0) to 1011(11) and back to zero when 1100 (12)).**

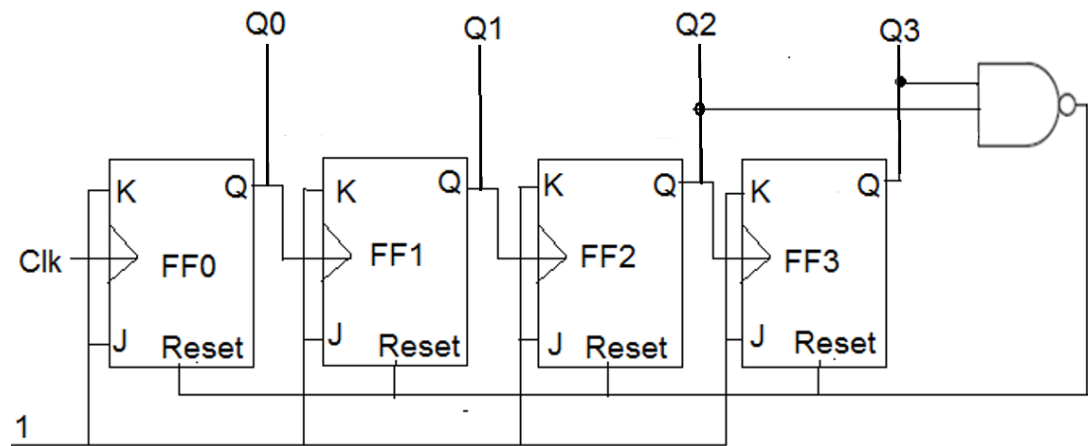


Figure 4.